

Entregables Cálculo

1. $f(x) = -\sin(x)$

Encuentro puntos de corte en intervalo:

$$-\sin(x) = 0 \rightarrow x = 0, \pi, 2\pi$$

Cálculo el Volumen y la Integral Asociada:

$$V = \pi \int_0^{\pi} (-\sin(x))^2 dx + \pi \int_{\pi}^{2\pi} (-\sin(x))^2 dx$$

$$V = \pi \int_0^{\pi} \sin^2(x) dx + \pi \int_{\pi}^{2\pi} \sin^2(x) dx$$

$$\int \sin^2(x) dx = \int \frac{1 - \cos(2x)}{2} dx \quad \cos^2(x) = \frac{1 + \cos(2x)}{2}$$

$$\int 1 - \frac{\cos(2x)}{2} dx = \int \frac{1 - \cos(2x)}{2} dx$$

$$= \frac{x}{2} - \frac{\cos(2x)}{4} + C$$

$$V = \pi \left[\frac{x}{2} - \frac{\cos(2x)}{4} \right]_0^{\pi} + \pi \left[\frac{x}{2} - \frac{\cos(2x)}{4} \right]_{\pi}^{2\pi}$$

$$V = \pi \left[\frac{\pi}{2} - \frac{\cos(2\pi)}{4} - \left(\frac{0}{2} - \frac{\cos(0)}{4} \right) \right] + \left[\frac{2\pi}{2} - \frac{\cos(4\pi)}{4} - \left(\frac{\pi}{2} - \frac{\cos(2\pi)}{4} \right) \right]$$

$$V = \pi \left(\frac{\pi}{2} - \frac{1}{4} + \frac{1}{4} \right) = \pi \cdot \frac{\pi}{2}$$

2. $\lambda(x) = \frac{dm}{dx} \rightarrow m = \int \lambda dx$

$$f(x) > 0 \quad \forall x \in D$$

Límites de Integración: 0 y 4

$$\int_0^4 2 + \frac{x}{2} + \frac{1}{\sqrt{x+1}} dx = \int_0^4 2 + \frac{x}{2} + \frac{1}{\sqrt{x+1}} dx$$

$$= \left[2x + \frac{x^2}{4} + 2\sqrt{x+1} \right]_0^4$$

$$m = \left(2(4) + \frac{4^2}{4} + 2\sqrt{4+1} \right) - \left(2(0) + \frac{0^2}{4} + 2\sqrt{0+1} \right)$$

$$m = 12 + 2\sqrt{5} - 2 = 10 + 2\sqrt{5} \text{ kg} = 14.47 \text{ kg}$$

3. Por Bolzano:

$$\exists x_0 \in (a, b) \mid \text{signo}(a) \neq \text{signo}(b) \mid f(x_0) = 0$$

Para que $\text{signo}(a) \neq \text{signo}(b)$: $\int_0^2 e^{t^2} dt > 1$ y

$a < b > 0$. Asumo que $b = 2$ por lo

$$f(0) = -1 + \int_0^0 e^{t^2} dt = 0 - 1$$

$$f(2) = -2 + \int_0^2 e^{t^2} dt = > 0$$

Asumo ^{que} cambio de signo entonces: $\exists x_0$

$$\exists \text{ al menos un } (x_0) \in [0, 2] \mid f(x_0) = 0$$

Para saber que es solo 1, encuentro la derivada y estudio el crecimiento en intervalo

Derivo Bajo en signo de Integral

$$f'(x) = f[g(x)]g'(x) - f[h(x)]h'(x) \quad \text{para } \int_{h(x)}^{g(x)} f(x)$$

$$f'(x) = e^{x^2} \cdot 1 - e^{0^2} \cdot 0 = e^{x^2}$$

$$f'(0) = 1$$

$$f'(2) = e^4 > 0$$

$f(x)$ es creciente $\forall x \in [0, 2]$ entonces:

$$\exists! (x_0) \neq f(x) \in [0, 2] \mid f(x_0) = 0$$

4. $\lim_{n \rightarrow \infty} \left(e^{\frac{1}{n}} \right)^{\frac{n^2+5}{n+1}}$

$$\lim_{n \rightarrow \infty} 1^{\frac{\infty}{\infty} \rightarrow \infty}$$

$$\lim_{n \rightarrow \infty} \left(e^{\frac{1}{n}} \right)^{\frac{n^2+5}{n+1}} = 1^{\infty} = \text{IND}$$

$$\lim_{n \rightarrow \infty} a_n = e^{\lim_{n \rightarrow \infty} \left(\frac{n^2+5}{n+1} \right) (e^{\frac{1}{n}} - 1)}$$

$$\lim_{n \rightarrow \infty} a_n = \frac{(n^2+5)(e^{\frac{1}{n}} - 1)}{n+1} = \frac{\infty}{\infty} \xrightarrow{\infty} \infty$$

$$\lim_{n \rightarrow \infty} \left(e^{\frac{1}{n}} \right)^{\frac{n^2+5}{n+1}} = \frac{\infty}{\infty} e^{\infty} = \infty$$

$$5. \lim_{n \rightarrow \infty} \frac{e^{a/2} + e^{a/2} + e^{a/2} + \dots + e^{a/n}}{\ln n} - n$$

Debido a las consecuencias del criterio de Stolz:

$$\lim_{n \rightarrow \infty} \frac{a_1 + a_2 + a_3 + \dots + a_n}{n} = \frac{1}{2}$$

Aplico la propiedad de suma y resta de límites:

Criterio de Stolz:

$$\lim_{n \rightarrow \infty} \frac{x_n}{y_n} = \lim_{n \rightarrow \infty} \frac{x_n - x_{n-1}}{y_n - y_{n-1}}$$

$$\lim_{n \rightarrow \infty} \frac{x_n}{y_n} = \lim_{n \rightarrow \infty} \frac{(e^{a/2} + e^{a/2} + e^{a/2} + \dots + e^{a/n}) - (e^{a/2} + e^{a/2} + e^{a/2} + \dots + e^{a/(n-1)})}{\ln n - \ln(n-1)}$$

$$\lim_{n \rightarrow \infty} \frac{e^{a/n}}{\ln(n) - \ln(n-1)} = \lim_{n \rightarrow \infty} \frac{e^{a/n}}{\ln(\frac{n}{n-1})} = e^{1/2}$$

$$\lim_{n \rightarrow \infty} \frac{n}{\ln(n)} = \infty$$

$$\lim_{n \rightarrow \infty} \frac{e^{a/2} + e^{a/2} + e^{a/2} + \dots + e^{a/n}}{\ln n} = \infty \cdot e^{1/2} - \infty = -\infty$$

$$6. \sum_{n=1}^{\infty} \frac{(n!)^2 (4^n)}{(2n)!}$$

Aplico Criterio de cociente

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = k$$

$$\lim_{n \rightarrow \infty} \frac{\frac{(n+1)!^2 (4^{n+1})}{(2n+2)!}}{\frac{(n!)^2 (4^n)}{(2n)!}} = \lim_{n \rightarrow \infty} \frac{(n+1)!^2 (4^{n+1}) (2n)!}{(n!)^2 (4^{n+1}) (2n+2)!}$$

$$= \lim_{n \rightarrow \infty} \frac{(n+1)!^2 (4^n) (4) (2n)!}{(n!)^2 (4^n) (2n+2)!} \quad (n+1)! = n! \cdot n+1$$

$$= \lim_{n \rightarrow \infty} \frac{(n+1)(n+1) (4^n) (4) (2n)!}{(n!)^2 (4^n) (2n+2)!} = \frac{4(n+1)^2 (2n)!}{(2n+2)!}$$

$$= \lim_{n \rightarrow \infty} \frac{4(n+1)^2}{2(n+1)} = 6 \quad \lim_{n \rightarrow \infty} \frac{4(n+1)}{2} = \infty$$